

QuorumProof — Whitepaper

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An agent fleet that reaches auditable decisions from Attestcoin-verified cross-chain facts. BUIDL CTC 2026 Fall · AI track · September 2026 · v1

1. Problem

Credit infrastructure is going autonomous. AI agents already draft credit decisions; they will soon execute them — releasing collateral, marking loans repaid, gating credit lines. Two blockers stand in the way:

1. **Unverified data.** An agent that reasons about a repayment on Ethereum while sitting on Creditcoin must trust a bridge or an oracle operator — centralized parties that can lie, lag, or get hacked. The Attestcoin Protocol eliminates exactly this: any source-chain transaction can be proven inside Creditcoin's trust domain via on-chain precompiles.
2. **Unaccountable judgment.** A single agent — even a good one — is one prompt injection, one buggy model update, one adversarial fact away from moving money wrongly. Finance never accepted "trust my sole judgment" from a human clerk; it will not accept it from software either. The gap is not better models. The gap is **process**: how a decision is reached, and whether it can be audited after the fact.

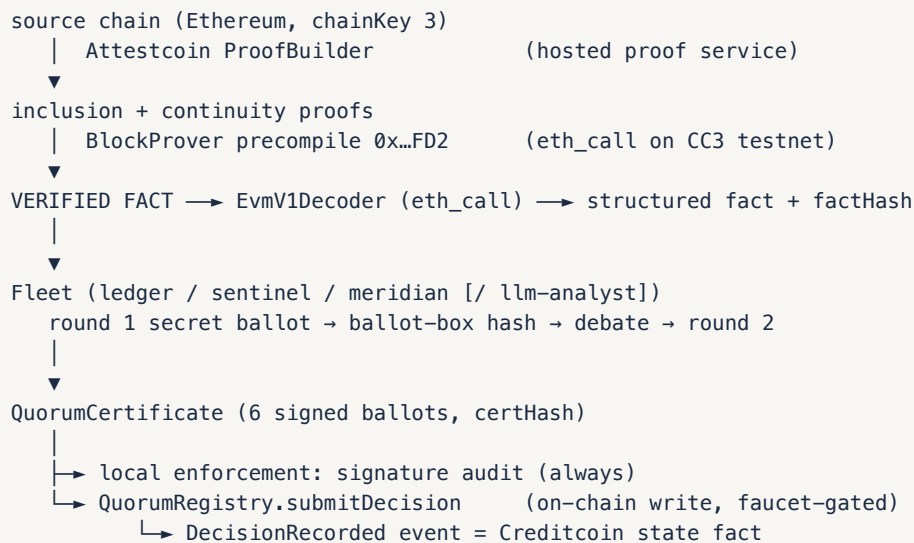
2. Idea

QuorumProof treats autonomous decisions the way consensus treats blocks.

A fleet of three heterogeneous agents — an underwriter (`ledger-01`), a fraud hunter (`sentinel-02`), a market analyst (`meridian-03`), plus an optional fourth LLM analyst — deliberates over facts that have already passed Attestcoin verification:

- **Round 1 — secret ballot.** Each agent independently votes APPROVE / DENY / ABSTAIN with a confidence and reasons. Ballots are committed into a ballot-box hash before anyone sees anyone's vote.
- **Round 2 — debate.** The tally is published. Each agent may revise exactly once under a *documented* revision rule (e.g., a sole dissenting approver keeps its vote but discounts confidence and states the jurisdictional reason on the record).
- **Quorum gate.** EXECUTE requires $\geq N$ approvals with sufficient mean confidence; otherwise REJECT. *Either way* the outcome is a signed **QuorumCertificate**: every ballot carries a secp256k1 signature bound (byte-exact, TS $\leftrightarrow$ Solidity) to the Attestcoin factHash, policyId, round, and vote.
- **Enforcement.** The certificate is audited locally (signature recovery) and, when configured, submitted to `QuorumRegistry` on CC3 — which recovers every ballot on-chain, applies the quorum gate a second time in Solidity, and emits `DecisionRecorded`. The decision becomes Creditcoin state that any lending protocol can consume.

3. Architecture



Trust properties:

- **No unverified data.** `attestTx` throws unless the BlockProver precompile returns true. The fleet literally cannot see unverified facts.
- **No single-model authority.** Agents differ in mandate *and* failure mode; the debate rules are public, deterministic, and printed into the audit trail.
- **No key ceremonies.** Verification is reads (`eth_call`); agent keys are ephemeral session keys whose signatures are self-certifying in the certificate.
- **Replayability.** `factHash` + `policyId` + `certHash` + 6 recoverable signatures = a decision a third party can re-derive end to end.

4. What it unlocks

- **Cross-chain repayment verification → autonomous collateral release** (the demo): a 3.6 ETH obligation is released when a quorum of agents verifies the actual mainnet repayment and blocked for a look-alike inflow.
- The same gate generalizes to DePIN settlement (uptime invoices), RWA income verification (creator finance), covenant enforcement — anywhere an autonomous action needs to be defensible *after the fact*.

5. Evaluation

Live, end-to-end, on public infrastructure (CC3 testnet; mainnet facts via the testnet's chainKey 3 registry):

- EXECUTE path: real 3.5999664 ETH repayment to the policy lender → 3/3 unanimous, quorum met, certificate signed by 3 distinct keys, audit clean.

- **REJECT** path: legitimate 0.0569 ETH transfer to a third party → Sentinel approves (integrity), Ledger and Meridian refuse (policy fit) → 1/3, quorum blocks; Sentinel revises with a documented sole-dissent statement; certificate still recorded — a *rejected* decision is as auditable as an executed one.
- Tamper test: flipping any signed vote breaks signature recovery (test suite asserts this).
- Full pipeline latency $\approx 2\text{--}3$ s per assessment (proof ≈ 1 s, verify ≈ 0.3 s, decode ≈ 0.3 s).

6. Roadmap

1. **Hackathon** → **CEIP**: deploy QuorumRegistry to CC3 testnet via the Discord faucet; re-run demo in on-chain-write mode.
2. **Fleet-as-a-service**: policy packs per vertical; agents as independent processes (today: in-process brains with identical interfaces — the seams for networked agents are already drawn).
3. **Heterogeneous model fleet**: route each brain to a different model provider (OpenAI-compatible adapter shipped) so no single vendor owns the quorum.
4. **Batch decisions**: `verifyBatch` (max 10 proofs / 1000 blocks per continuity proof) for high-volume settlement.

7. Limitations (stated plainly)

Default brains are deterministic evaluators — chosen for auditability and a zero-cost, keyless demo; the LLM adapter exists but is off unless configured. QuorumRegistry is not yet deployed (faucet-gated); local enforcement is complete and labeled. Ephemeral agent keys demonstrate signature-bound accountability; a production roster would register persistent control addresses in `agentControl` (the contract supports this today).